The documentation and process conversion measures necessary to comply with this revision shall be completed by 25 February 1995

INCH-POUND

MIL-S-19500/156J
25 November 1994
SUPERSEDING
MIL-S-19500/156H
23 November 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE
TYPES 1N935B, 1N937B, 1N938B, 1N939B, 1N940B, AND 1N935B-1, 1N937B-1,
1N938B-1, 1N939B-1, AND 1N940B-1, 1N935BUR-1, 1N937BUR-1, 1N938BUR-1,
1N939BUR-1, AND 1N940BUR-1, JAN, JANTX, JANTXV AND JANS;
RADIATION HARDENED (TOTAL DOSE ONLY)
TYPES JANTXVM, D, L, R, F, G, H, AND JANSM, D, L, R, F, G, H

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 9.0 volts ±5 percent, silicon, voltage-reference diodes. Four levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500. Seven levels of radiation hardened (total dose only) product assurance are provided for each encapsulated device type as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1 (DO-7 and DO-35) and figure 2 (DO-213AA).

1.3 Unless otherwise specified, maximum ratings (T_A = +25°C).

P_T 1/	T_{STG} and T_{op}	I_{ZM}	Power derating above T_A = +25°C
mW	°C	mA dc	mW/°C
500	-65 to +175	50	3.33

1/ For zero-TC voltage reference operation, I_Z is limited to 7.5 mA dc over -55°C to +150°C.

1.4 Primary electrical characteristics. Unless otherwise specified, primary electrical characteristics at T_A = +25°C.

Type	ΔV_Z (voltage - temperature stability) I_Z = 7.5 mA dc	Z_Z I_Z = 7.5 mA dc	V_Z I_Z = 7.5 mA dc		I_R V_R = 6.0 V
			Min	Max	
	mV dc	ohms	volts	volts	µA
1N935B, 1N935B-1, 1N935BUR-1	184.0	20	8.55	9.45	10.0
1N937B, 1N937B-1, 1N937BUR-1	37.0	20	8.55	9.45	10.0
1N938B, 1N938B-1, 1N938BUR-1	19.0	20	8.55	9.45	10.0
1N939B, 1N939B-1, 1N939BUR-1	9.0	20	8.55	9.45	10.0
1N940B, 1N940B-1, 1N940BUR-1	3.7	20	8.55	9.45	10.0

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Services Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

UR - - - - - - - - - - - - - - Unleaded or surface mounted (round endcapped diodes).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

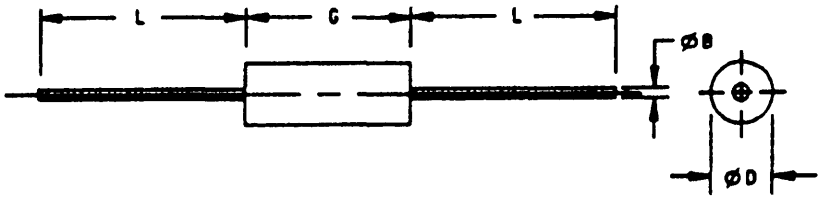
3.3.3 Dash-one construction. These devices shall be category I or II metallurgical bonding in accordance with MIL-S-19500.

3.3.4 JANS construction. Construction shall be dash-one, category I or II metallurgical bond in accordance with MIL-S-19500, appendix A, 30.14.3 and 30.14.4.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of "UR" version devices. For "UR" version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

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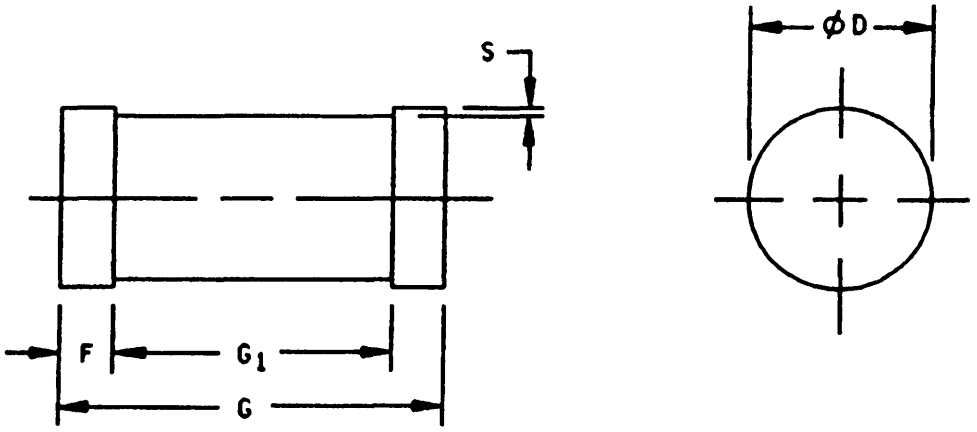


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.018	.023	0.46	0.58	3,4
ϕD	.060	.107	1.52	2.72	5
G	.120	.300	3.05	7.62	
L	1.000	1.500	25.40	38.10	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The specified lead diameter applied in the zone between .050 inch (1.27 mm) and 1.0 inch (25.40 mm) from the diode body. Outside of this zone the lead diameter is not controlled.
4. Both leads shall be within the specified dimension.
5. The minimum body diameter shall be maintained over .15 inch (3.8 mm) of body length.

FIGURE 1. Physical dimensions, 1N935B, 1N937B, 1N938B, 1N939B, 1N940B, 1N935B-1, 1N937B-1, 1N938B-1, 1N939B-1, and 1N940B-1 (DO-35 and DO-7).

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
ϕD	.063	.067	1.60	1.70
F	.016	.022	0.41	0.56
G	.130	.146	3.30	3.70
G_1	.100 ref		2.54 ref	
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions, 1N935BUR-1, 1N937BUR-1, 1N938BUR-1, 1N939BUR-1 and 1N940BUR-1 (DO-213AA).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein except lot accumulation requirement shall be six months in lieu of six weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and herein.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table III herein.

4.2.2 Radiation hardened devices. See MIL-S-19500 and 4.4.4 herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
11	V_Z, Z_Z, and I_R	V_Z, Z_Z, and I_R
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; $\Delta Z_Z \leq \pm 15$ percent of initial reading At T_A = +25°C, ±2°C, $\Delta V_Z \leq \pm 4$ mV dc from initial value for 1N935B, 1N935B-1, 1N935BUR-1, 1N937B, 1N937B-1, and 1N937BUR-1. $\Delta V_Z \leq \pm 3$ mV dc from initial value for 1N938B, 1N938B-1, 1N938BUR-1, 1N939B, 1N939B-1 and 1N939BUR-1. $\Delta V_Z \leq \pm 2$ mV dc from initial value for 1N940B, 1N940B-1, and 1N940BUR-1.	Subgroup 2 of table I herein $\Delta Z_Z \leq \pm 15$ percent of initial reading At T_A = +25°C, ±2°C, $\Delta V_Z \leq \pm 4$ mV dc from initial value for 1N935B, 1N935-1, 1N935BUR-1, 1N937B, 1N937B-1, and 1N937BUR-1. $\Delta V_Z \leq \pm 3$ mV dc from initial value for 1N938B, 1N938B-1, 1N938BUR-1, 1N939B, 1N939B-1, and 1N939BUR-1. $\Delta V_Z \leq \pm 2$ mV dc from initial value for 1N940B, 1N940B-1, and 1N940BUR-1.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

I_Z = 7.5 mA dc, +0.75 mA dc, -0 mA dc, T_A = +150°C +5°C, -0°C.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein.

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4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B3	1071	Test condition E.
B4	1037	I_Z = 7.5 mA dc at T_A = +150°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No forced air cooling on the device shall be permitted.
B5	1027	I_Z = 7.5 mA dc for 1,000 hours. T_A = +150°C.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
B2	1071	Test condition E.
B3	1027	T_A = +150°C and I_Z = 7.5 mA dc. NOTE: When group C (subgroup 6) is required, group B (subgroup 3) is not required for that lot.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Lead tension: 4 pounds weight, t = 15 ±3 seconds; test condition A. Not applicable to "UR" version devices. Lead fatigue: Test condition E, not applicable to "UR" version devices.
C2	1071	Test condition E.
C6	1026	T_A = +100°C and I_Z = 7.5 mA dc (see 4.5.2).

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with MIL-S-19500 and table II herein. MIL-S-19500, appendix G is not required when homogeneous product of radiation hard assembly (in accordance with applicable 360 form) is submitted and successfully passes group D sampling of total dose at an outside facility compliant to MIL-STD-750, method 1019. Submitted lots for group D sample inspection must be constructed using one homogeneous wafer lot for the zener and one wafer lot for the compensating die (die), as also described in the submitted 360 forms.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage-temperature stability. The breakdown voltage of each diode type shall be measured and recorded at each of the specified temperatures. The lowest measured voltage shall be subtracted from the highest measured voltage for each diode. The difference value obtained shall not exceed the specified ΔV_Z per diode type at I_Z = 7.50 ±0.01 mA dc.

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4.5.2 Reference voltage time stability. The breakdown voltage shall be measured prior to life testing, at 340 hours, and at the conclusion of the life test as shown in table below. The 340-hour reading shall be compared with the 0-hour reading and the 1,000-hour reading compared with the 340-hour reading. The change in breakdown voltage shall not exceed the limits specified.

Inspection	MIL-STD-750 conditions	Symbol	Limits		Unit
			Min	Max	
Reference-voltage time stability	T_A = +100°C ±2°C (see 4.5.2 and 4.5.3) I_Z = 7.5 ±0.01 mA dc	ΔV_Z			
1N935B, 1N935B-1, 1N935BUR-1	(0 to 340 hours)			7	mV dc
1N937B, 1N937B-1, 1N937BUR-1				7	mV dc
1N938B, 1N938B-1, 1N938BUR-1				6	mV dc
1N939B, 1N939B-1, 1N939BUR-1				5	mV dc
1N940B, 1N940B-1, 1N940BUR-1				3	mV dc
1N935B, 1N935B-1, 1N935BUR-1	(340 to 1,000 hours)			4	mV dc
1N937B, 1N937B-1, 1N937BUR-1				4	mV dc
1N938B, 1N938B-1, 1N938BUR-1				3	mV dc
1N939B, 1N939B-1, 1N939BUR-1				3	mV dc
1N940B, 1N940B-1, 1N940BUR-1				1	mV dc

4.5.3 Reference voltage. The test current shall be applied until thermal equilibrium is attained (15 seconds minimum) prior to reading the reference voltage. For this test, the diode shall be suspended by its leads with mounting clips whose inside edge is located between .375 inch (9.53 mm) and .500 inch (12.70 mm) inch from the body and the mounting clips shall be maintained at the specified temperature. This measurement may be performed after a shorter time following application of the test current than that which provides thermal equilibrium if correlation to stabilized readings can be established to the satisfaction of the Government.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reference voltage (see 4.5.3)	4022	I_Z = 7.5 ±0.01 mA dc	V_Z	8.55	9.45	V dc
Small-signal breakdown impedance	4051	I_Z = 7.5 ±0.1 mA dc I_{sig} = 0.75 mA ac	Z_Z		20.0	Ω
Reverse current leakage	4016	DC method; V_R = 6.0 V dc	I_R		10.0	µA
Subgroup 3						
Voltage-temperature stability (see 4.5.1 and 4.5.3)		I_Z = 7.5 ±0.01 mA dc T_A = -55°C, 0°C, +25°C, +75°C, +150°C ±2°C	ΔV_Z			
1N935B, 1N935B-1, 1N935BUR-1					184.0	mV dc
1N937B, 1N937B-1, 1N937BUR-1					37.0	mV dc
1N938B, 1N938B-1, 1N938BUR-1					19.0	mV dc
1N939B, 1N939B-1, 1N939BUR-1					9.0	mV dc
1N940B, 1N940B-1, 1N940BUR-1					3.7	mV dc
Subgroups 4, 5, and 6						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		ΔV_{ZS} Pre- or post-irradiation change		Unit
	Method	Conditions		M, D, R, L, F, G, and H		M, D, R, L, F, G, and H				
				Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = +25°C								
Steady-state total dose irradiation	1019	I_Z = 7.5 ±0.01 mA dc, condition A								
Reference voltage (see 4.5.3)	4022	I_Z = 7.5 ±0.01 mA dc	V_Z	8.55	9.45	8.55	9.45			V dc
Small-signal breakdown impedance	4051	I_Z = 7.5 ±0.01 mA dc, I_{sig} = 0.75 ac	Z_Z		20		20			ohms
Reverse current leakage	4016	DC method, V_R = 6.0 V dc	I_R		10.0		10.0			µA
Voltage stability (see 4.5.3)	4022	I_Z = 7.5 ±0.01 mA dc, T_A = +25°C ±2°C	ΔV_Z							
1N935B, 1N935B-1 1N935BUR-1									±4.0	mV
1N937B, 1N937B-1 1N937BUR-1									±3.5	mV
1N938B, 1N938B-1 1N938BUR-1									±3.0	mV
1N939B, 1N939B-1 1N939BUR-1									±2.5	mV
1N940B, 1N940B-1 1N940BUR-1									±1.5	mV

1/ For sampling plan, see MIL-S-19500.
2/ Group D qualification may be performed anytime prior to lot formation.

TABLE III. Group E inspection (all quality levels) for qualification only.

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Inspection	MIL-STD-750		Sampling plan 1/
	Method	Conditions	
Subgroup 1			22 devices
Thermal shock temperature cycling	1051	500 cycles	c = 0
Electrical measurement		See table 1, subgroup 2	
Subgroup 2			22 devices
Steady-state dc life	1038	Condition B; 1,000 hours, I_Z = 7.5 mA dc, T_A = +150°C	c = 0
Electrical measurement		See table 1, subgroup 2	
Subgroup 3			
Not applicable			
Subgroups 4, 5, and 6			
Not applicable			

1/ For sampling plans not specified, see MIL-S-19500.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.1).

c. Product assurance level and type designator.

6.4 Substitution of radiation hardened devices. See MIL-S-19500.

6.5 Substitution of devices. Device types within this series with higher type numbers (lower ΔV_Z voltage-temperature stability) are a direct oneway substitution for lower type numbers (higher ΔV_Z voltage-temperature stability).

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, SM
Navy - AS, CG, MC
Air Force - 19, 80, 85, 99

Preparing activity:
DLA - ES

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/156J	2. DOCUMENT DATE (YYMMDD) 941125

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, VOLTAGE-REFERENCE TYPES 1N935B, 1N937B, 1N939B, 1N940B, AND 1N935B-1, 1N937B-1, 1N938B-1, 1N939B-1, AND 1N940B-1, 1N935BUR-1, 1N937BUR-1, 1N938BUR-1, 1N939BUR-1, AND 1N940BUR-1, JAN, JANTX, JANTXV, AND JANS; RADIATION HARDENED (TOTAL DOSE ONLY) TYPES JANTXVM, D, L, R, F, G, H, AND JANSM, D, L, R, R, F, G, AND H

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (if applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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